

Communication Latency in Distributed Inference

PROGRESS - THESIS FOLLOW-UP - OCTOBER 14, 2026

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Status (week #1)

- Read PICO (the awarded ISC-HPC conference paper)
  - Framework for benchmarking collective operations with step-level profiling and run-metadata records for reproducibility
- Models for spike:
  - Llama 3.1 405B – biggest open dense transformer that can fit on B200 node
  - Llama 3.1 70B – if time allows for TP scaling and KV saturation
- Spike access:
  - Contacted Hengjian, can use Spike as soon as I see available GPUs
  - Will schedule with him once experiments are set up and ready

Planning the Paper Scope

Main RQ: How does communication latency and interconnect quality affect tensor-parallel inference performance, scaling efficiency, and energy consumption?

- **Keep:** TP scaling (fold in KV cache saturation), latency injection curves, SLO tolerance, energy per token
- **Leave out:** single GPU, V100, hardware comparison
- **Adding:** power over time, B200 latency injection with large model, nsys AllReduce duration and counts (compare to nccl-tests)

Next steps (next two weeks)

- Add `nsys` measurements of AllReduce counts and durations to the experiment scripts
- Test experiments on Spike if compute available
  - Check if everything works like latency injection
- Solidify how paper's story is set up, begin to collapse KV cache saturation into TP scaling portion

Questions

- A100 data? Plan is to add repeated runs with nsys or anything else?
- Contract?

Proposed Timeline

Week	Date	Focus
1	Oct 12	Spike test, experiment scripts, nsys setup
2	Oct 19	nsys profiling on A100, repeated runs
3	Oct 26	
4	Nov 2	Refine AllReduce model with nsys data
5	Nov 9	B200 experiments (405B at TP=8, latency injection, 70B TP scaling)
6	Nov 16	
7	Nov 23	Full draft, ready for feedback
8	Nov 30	Revised final draft incorporating comments

Thank you!